

1. (10%) What is the maximum number of nodes in a k -ary tree of height h ? Prove your answer.

$$k=2, h=3 \text{ 時: } 2^0 + 2^1 + 2^2 = 7$$

$$k=3, h=3 \text{ 時: } 3^0 + 3^1 + 3^2 = 13$$

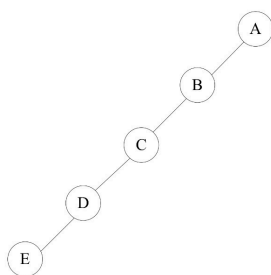
$$k=i, h=n \text{ 時: } i^0 + i^1 + i^2 + \dots + i^{n-1}$$

$\rightarrow a_1=1, r=k, \text{項數(樹高)}=h \text{ 的等比級數}$

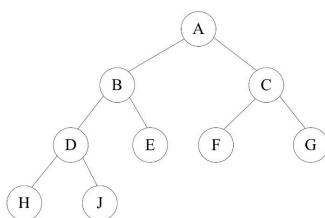
$$S_n = \frac{1 \times (k^h - 1)}{k - 1} = \frac{k^h - 1}{k - 1}$$

$$\text{Ans} = \frac{k^h - 1}{k - 1} \text{ 個}$$

2. (20%) Write out the inorder, preorder, postorder, and level-order travels for the binary trees (a) and (b).



(a)



(b)

(a)

inorder: $E \rightarrow D \rightarrow C \rightarrow B \rightarrow A$

preorder: $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$

postorder: $E \rightarrow D \rightarrow C \rightarrow B \rightarrow A$

level-order: $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$

(b)

inorder: $H \rightarrow D \rightarrow J \rightarrow B \rightarrow E \rightarrow A \rightarrow F \rightarrow$

$C \rightarrow G$

preorder: $A \rightarrow B \rightarrow D \rightarrow H \rightarrow J \rightarrow E \rightarrow C \rightarrow$

$F \rightarrow G$

postorder: $H \rightarrow J \rightarrow D \rightarrow E \rightarrow B \rightarrow F \rightarrow G \rightarrow$

$C \rightarrow A$

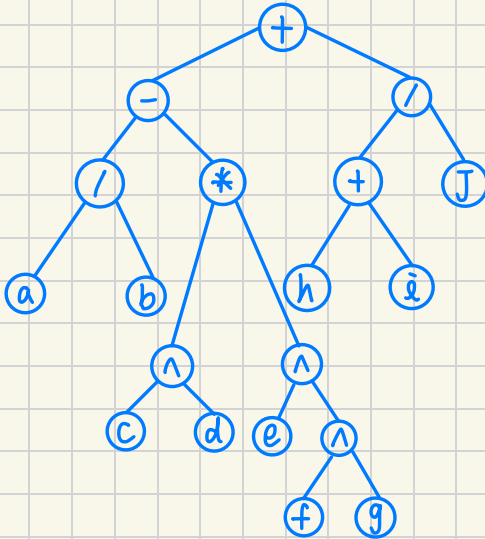
level-order: $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G \rightarrow$

$H \rightarrow J$

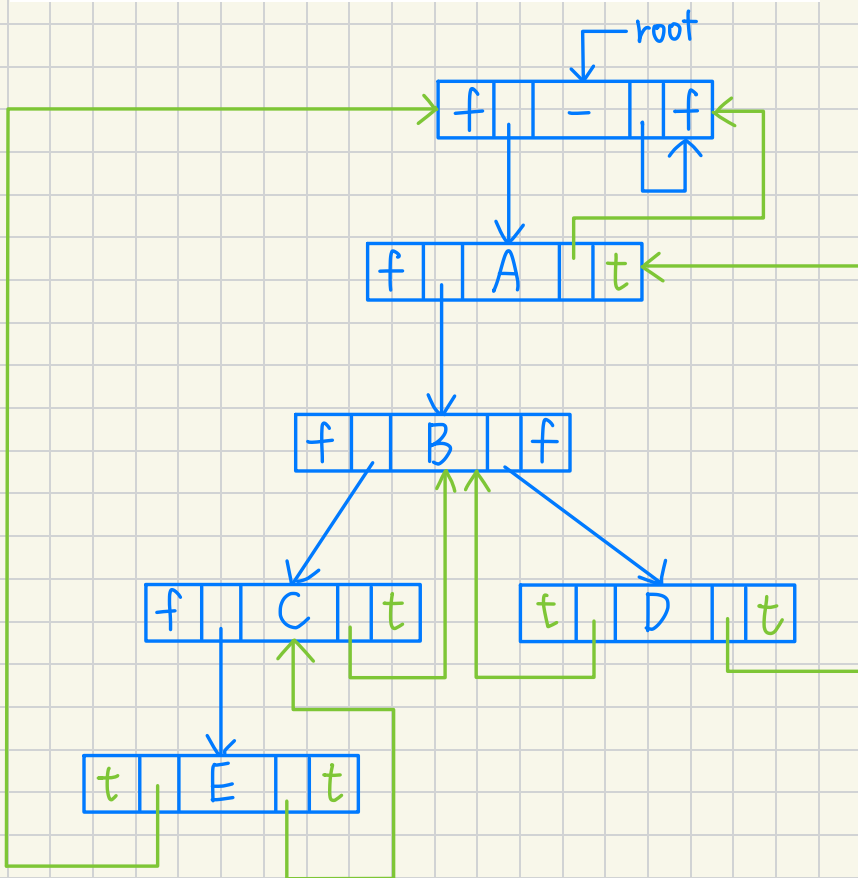
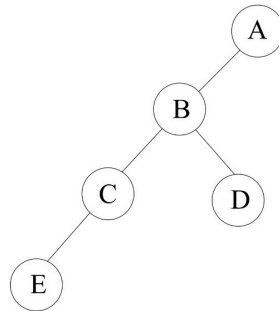
3. (15%) Considering operator precedence, draw the expression tree of the infix expression. Note that $\wedge$ is power.

$$\underline{a / b} - \underline{c \wedge d} * \underline{e \wedge f \wedge g} + \underline{(h + i) / j}$$

$$\rightarrow [a / b] - [c \wedge d] * [e \wedge f \wedge g] + [(h + i) / j]$$



4. (10%) Draw the threaded representation of the following binary tree.



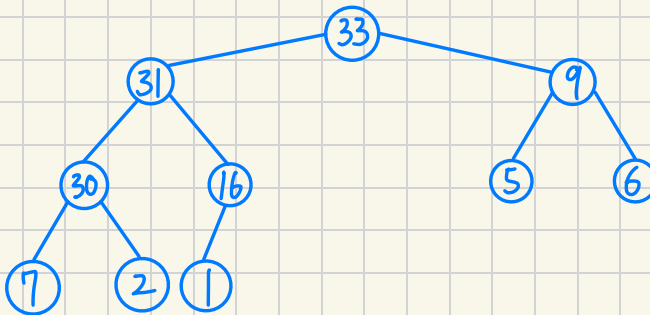
5. (10%) Suppose that we execute the following instructions:

**push 7, push 9, push 16, push 30, push 49, pop, push 82, push 5, push 33,
push 31, push 6, pop, push 2, push 1.**

- (a) What is the resultant heap if we use a max heap?
(b) What is the resultant heap if we use a min heap?

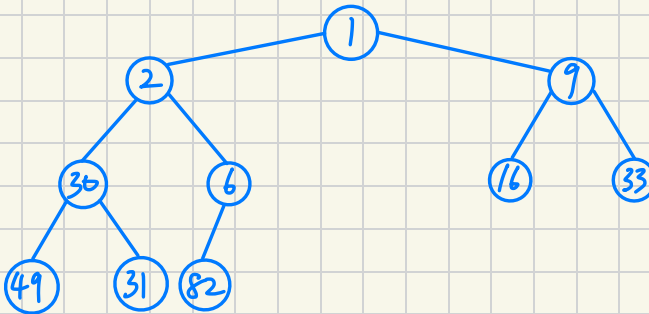
(a) max heap

PUSH 7	7
PUSH 9	9 7
PUSH 16	16 7 9
PUSH 30	30 16 9 7
PUSH 49	49 30 9 7 16
POP	30 16 9 7
PUSH 82	82 30 9 7 16
PUSH 5	82 30 9 7 16 5
PUSH 33	82 30 33 7 16 5 9
PUSH 31	82 31 33 30 16 5 9 7
PUSH 6	82 31 33 30 16 5 9 7 6
POP	33 31 9 30 16 5 6 7
PUSH 2	33 31 9 30 16 5 6 7 2
PUSH 1	33 31 9 30 16 5 6 7 2 1



(b) min heap

PUSH 7	7
PUSH 9	7 9
PUSH 16	7 9 16
PUSH 30	7 9 16 30
PUSH 49	7 9 16 30 49
POP	9 30 16 49
PUSH 82	9 30 16 49 82
PUSH 5	5 30 9 49 82 16
PUSH 33	5 30 9 49 82 16 33
PUSH 31	5 30 9 31 82 16 33 49
PUSH 6	5 6 9 30 82 16 33 49 31
POP	6 30 9 31 82 16 33 49
PUSH 2	2 6 9 30 82 16 33 49 31
PUSH 1	1 2 9 30 6 16 33 49 31 82

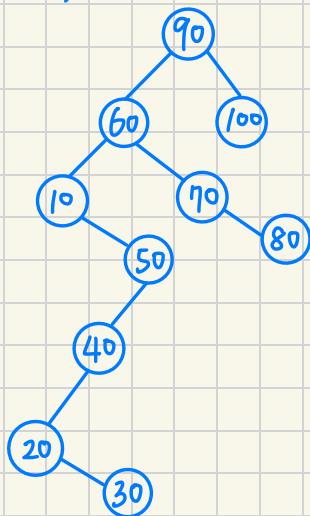


6. (10%) Given the following numbers:

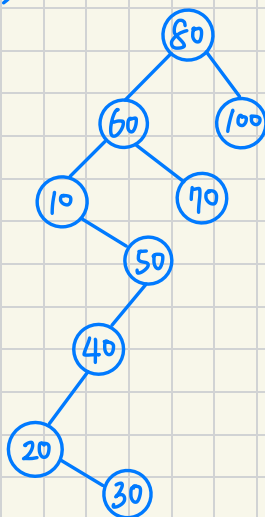
90, 60, 70, 100, 10, 50, 40, 20, 30, 80.

- Construct a binary search tree.
- From (a), what does the binary search tree become if we delete 90 using the rule "replaced by the largest element in its left subtree"?

(a)



(b)



7. (12%) How many distinct 6-node binary tree structures can we create?

$$\frac{12!}{7!6!} = \frac{12 \times 11 \times 10 \times 9 \times 8}{6 \times 5 \times 4 \times 3 \times 2 \times 1} = 12 \times 11 = 132 \neq$$

8. (13%) Transforming the forest into a binary tree using the rule:

- If $T_1, \dots, T_n$ is a forest of trees, then the binary tree corresponding to this forest, denoted by $B(T_1, \dots, T_n)$
- Root equal to $\text{root}(T_1)$
- Left subtree equal to $B(T_{11}, T_{12}, \dots, T_{1m})$, where $T_{11}, \dots, T_{1m}$ are the subtrees of root (T_1); and has right subtree $B(T_2, \dots, T_n)$

